Chapter 2 Review

December 24, 2023

This is a document to review a couple concepts of FOL.

Why FOL-tableau would not work if we used a domain other than the domain of the denumerable parameters

This fact can be exhibited by looking back at Problem 2 of Chapter 9 of the Beginners Guide [Smullyan, 2014]. The question is the prove that if I satisfies a set S, then for any $\delta \in S$ and any parameter a that is new to S, the set $S \bigcup \delta(a)$ is satisfied by some extension. Because a is a parameter, this is really easy: the way parameters were defined allowed us to assign to them *any* element of the domain under any interpretation whatsoever; this is why this works for the parameters. However, if we used some other domain, given that I satisfies S, we don't even know that the elements of that domain are in the domain under I. And even if we did, we still don't know if $\delta(a)$ would be true under some extension of I, as a is no longer a parameter.

Clarifying the role of predicates in FOL-valuations

Recall that we have the following definition included in the definition of a FOL-valuation (in the domain of the parameters):
B_1: The sentence $\sim X$ receives the value t under v iff $v(X) = f$; and accordingly $v(\sim X) = f$ iff $v(X) = t$.
where X is a sentence. This means that we can still assign t or f to predicates, as they are sentences (as long as its set of n-tuples $(a_1, ..., a_n)$ does not include any variables). However, we no longer have the notion of an interpretation $I(P)$

over a predicate P. But this is fine, as we can show that given any interpreation I, there is one and only one valuation v such that, for every sentence X, X is true under I iff X is true under v. This is just the valuation that assigns t to those and only those sentences that are true under I. Also, we can show that, for any valuation v, there is one and only one interpreation I such that, for every sentence X, X is true under I iff X is true under v. This is just the interpretation such that, for any predicate P of degree n, take $I(P)$ to be the relation that $P_{a_1,\ldots,a_n} = t$ under v.